

Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: ARABICMLLU | **Context:** Non-Arab Tourists and Expats in Eight Arabic-Speaking Countries (ArabicMLLU Deployment)

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's taxonomy of 40 tasks across school subjects and education levels [Q25] aligns well with the deployment's school-curriculum knowledge scope (history, geography, Arabic language, civics) [Q72, Q73]. However, several taxonomic issues materially harm content validity for this deployment: (a) the taxonomy includes categories irrelevant to the tourist/expat school-knowledge use case — Driving Test, professional Law (Morocco-only), vocational General Knowledge — that were observed in the sampled data [DATASET-D5, DATASET-D26, DATASET-D27]; (b) university-level content is only 6% of the dataset [Q71]; (c) Civics content as instantiated is overwhelmingly Jordanian-specific rather than pan-Arab [DATASET-D4]; (d) Morocco's school-curriculum (history/geography/civics) is essentially absent from the sample, with Morocco's contribution concentrated in professional law [DATASET-D5]; (e) Kuwait does not appear at all in the sample [Limitations.5]. The future-work acknowledgment of needed extensions — short-answer questions, larger regional coverage, professional-domain content [Q88] — confirms these are recognized gaps.

ID	Evidence
Q25	"ArabicMLLU is an Arabic multiple-choice question-answering dataset comprising 40 tasks spanning a wide range of subjects and education levels." (p.3)
Q71	"This could be attributed to the relatively small portion (i.e., 6%) of university-level questions in ArabicMMU, which potentially skews the results." (p.7)
Q72	"We present the performance of open-source models on selected subjects that potentially contain Arabic-specific contexts." (p.7)
Q73	"These subjects include history, geography, civics, political" (p.7)
Q88	"For future work, ArabicMLLU can be extended to include short-answer or essay questions, different modalities (i.e., images, audio, video), larger region coverage, and more questions in professional settings." (p.9)
D4	لُدُع...يندرأل ايب رعل اشي جلا تالايكشت نم في ربع العوقل... 1952م اندرأل ايف فيفالتئا ةموكح لوأ ليكشت مت All 5 sampled Civics Middle School items are exclusively about Jordanian political history
D5	ين طول اعافد اب ةفل كم الم ةي موكح الح اطلسل او لدعل ريزو نم... فانئتسالا لجأ ئدتبي (Professional) items are from Morocco (Moroccan procedural law)
D26	رتم 50 دعبلع؟ ععضو كليلع نيأ، سكاغال ريذحتل ثلثم كي دلو كتبكرم ل طعت لاح في Driving test questions from UAE, Lebanon, Egypt — practical/regulatory, not school-curriculum history/geography
D27	هتياهن نم بولسم لكشلا يبولول رامسم: وه جاصل انس يغرب Jordan primary GK: sheet metal screw definition — vocational/practical knowledge

Strengths

- Authentic school-curriculum framing across primary/middle/high school directly matches the deployment's knowledge scope [Q25, Q28, DATASET-D38, DATASET-D34]
- Subject groups explicitly include Arabic-specific contexts (history, geography, civics, political science) that are deployment-relevant [Q72, Q73]
- Broad multi-subject taxonomy with level-appropriate complexity confirmed in sampled data [DATASET-D38, DATASET-D22, DATASET-D34]

ID	Evidence
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Q25	“ArabicMMLU is an Arabic multiple-choice question-answering dataset comprising 40 tasks spanning a wide range of subjects and education levels.” (p.3)
Q28	“The subjects are drawn from different education levels (primary school, middle school, and KSA, prioritize Islamic studies alongside subjects like mathematics, natural science, social science, and geography.” (p.3)
Q72	“We present the performance of open-source models on selected subjects that potentially contain Arabic-specific contexts.” (p.7)
Q73	“These subjects include history, geography, civics, political” (p.7)
D22	قضية: مواب عملة لى غش التال عملة نإف... عملة لس تال خدملة تناك اذا هنأ بساحل ل و ح عة ج ا ر م ل ا ب و ل س أ ص ر ت ف ي — Egypt university accounting — auditing/financial statements questions
D34	— Jordan primary history: ancient name of Medina — Islamic history, factual ل م د ق ل م س ا ل و ه ب ر ث ي
D38	+ م و ي س ي ن غ ا م ل ا ؟ ل ص ح ي ذ ل ا ل ع ا ف ت ل ا ف ص ت ل ا ل ا ت ل ا ت ا ل د ا ع م ل ا ي ؛ ر ب ت خ م ل ا ب م و ي س ي ن غ ا م ل ا ن م ي م ك ق ر ح ب د م ح أ م ا ق — Natural science MS: chemistry reactions — country-neutral STEM م و ي س ي ن غ ا م ل ا د ي س ك أ -> ن ي ج س ك أ

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories for this deployment: Arabic language, Arab history, Arab geography, civics/social science at school-curriculum level [elicitation Q1, A1]; explicitly out of scope are dialectal phrases, tourist etiquette, practical travel, and professional domains.

IO-2: Yes — country-specific school-curriculum coverage is unequal. Morocco's school subjects are largely absent from the sample (only professional law observed) [DATASET-D5]; Kuwait does not appear at all [Limitations.5]; civics is Jordan-only in the sample [DATASET-D4]. The paper acknowledges unequal country representation [Q89, Q90, Q91].

IO-3: Yes — Driving Test [DATASET-D26], professional Law [DATASET-D5], vocational General Knowledge (sheet metal screws, hospital names) [DATASET-D27, DATASET-D10], and University Accounting/Management [DATASET-D22, DATASET-D23] are out of scope for the deployment's school-curriculum tourist knowledge use case.

IO-4: Documented gaps: (1) absence of Maghrebi school-curriculum content for Morocco; (2) absence of Kuwaiti curriculum; (3) lack of a multi-perspective or contested-content taxonomic dimension despite the subjects most relevant to the deployment being inherently contested [Q72, Q73]; (4) university-level content underrepresented at 6% [Q71].

ID	Evidence
Q1	“we present ArabicMMLU, the first multi-task language understanding benchmark for the Arabic language, sourced from school exams across diverse educational levels in different countries spanning North Africa, the Levant, and the Gulf regions.” (p.1)
Q25	“ArabicMMLU is an Arabic multiple-choice question-answering dataset comprising 40 tasks spanning a wide range of subjects and education levels.” (p.3)
Q71	“This could be attributed to the relatively small portion (i.e., 6%) of university-level questions in ArabicMMU, which potentially skews the results.” (p.7)
Q72	“We present the performance of open-source models on selected subjects that potentially contain Arabic-specific contexts.” (p.7)
Q73	“These subjects include history, geography, civics, political” (p.7)
Q88	“For future work, ArabicMMLU can be extended to include short-answer or essay questions, different modalities (i.e., images, audio, video), larger region coverage, and more questions in professional settings.” (p.9)
Q89	“ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all.” (p.9)

Q90	"This is largely due to the availability of publicly-accessible exams in each country; some countries have digitized their exams, but not others." (p.9)
Q91	"Additionally, our search for relevant Arabic content across the internet was not exhaustive." (p.9)
D4	ل دُع... يندرأل ايب رعل ا شيجل ا تال ي كشت ن م عي رعل ا ةوقل ا... 1952 ماع ندرأل ا ي ف ةيفال تئ ا ةموكح لوأ ل ي كشت م All 5 sampled Civics Middle School items are exclusively about Jordanian political history
D5	ي ن طول ا عاف دل اب ةفل كم الم ا مي موكح ل ا طلس ل او لدعل ا ريزو ن م... فانئ سالا ل ج أ ئ دب ي (Professional) items are from Morocco (Moroccan procedural law)
D10	Jordan — ةي بطل ا ةني دمل ا ي ف شت سم... ةص صخ ت م ل ا ساس ؤم ل ن م دي دعل ا مي موكح ل ا مي كل م ل ا مي بطل ا تام دخل ا لم شت Jordan primary social science: Jordan Royal Medical Services institutions
D22	ةي ع ب ت ل اب ةمي لس نوكت ل ي غ ش ت ل ا ةي لم ع ن ا ف... ةمي لس تال خ دم ل ا ت ن ا ك ا ذ ا ه ن ا ب س ا ح ل ا ل و ح ة ع ج ا ر م ل ا بول س ا ض ر ت ف ي Egypt university accounting — auditing/financial statements questions
D23	Severely garbled OCR text making question almost illegible — ة ز ك ر م ل ا : ي ل ا ر ي ت ذ ت... ص ا خ ت ل ا ل ا ن ي ن ب ط ل ت د ل ا ع ت ي ي ه ت ة ت ج ر د
D26	Driving test questions from UAE, Lebanon, Egypt — ر ت م 50 د ع ب ي ل ع ؟ ه ع ض و ك ي ل ع ن ي ا ، س ك ا ع ل ا ر ي ذ ح ت ل ا ث ل ث م ك ي د ل و ك ت ب ك ر م ل ط ع ت ل ا ح ي ف practical/regulatory, not school-curriculum history/geography
D27	Jordan primary GK: sheet metal screw definition — ه ت ي ا ه ن م بول سم لكش ل ا ي بول ول رام سم : وه جاصل ا نس ي غ ر ب vocational/practical knowledge
WEB-3	Palestine confirmed top-3 source country — yet no Palestinian annotators in pool (https://github.com/mbzuai-nlp/ArabicMMLU)

Information Gaps

- Exact per-country, per-subject distribution counts not extractable from documentation
- Whether Country=null items are truly pan-Arab or reflect missing metadata

Requires Expert Verification

- Whether sub-regional curriculum distinctions (Maghrebi, Levantine, Gulf) are meaningfully represented in the full dataset beyond the sampled configs

Remediation

Gap 1: Out-of-scope categories (Driving Test, professional Law, vocational General Knowledge) inflate aggregate scores irrelevantly [DATASET-D5, DATASET-D26, DATASET-D27]

Recommendation: Define a deployment-relevant subset of ArabicMMLU configs (history, geography, civics, Arabic language, social science, Islamic studies at appropriate framing) and report scores only on this subset; exclude driving test, professional law, and vocational items.

ID	Evidence
D5	All 5 sampled Law (Professional) items are from Morocco (Moroccan procedural law) — ي ن طول ا عاف دل اب ةفل كم الم ا مي موكح ل ا طلس ل او لدعل ا ريزو ن م... فانئ سالا ل ج أ ئ دب ي
D26	Driving test questions from UAE, Lebanon, Egypt — ر ت م 50 د ع ب ي ل ع ؟ ه ع ض و ك ي ل ع ن ي ا ، س ك ا ع ل ا ر ي ذ ح ت ل ا ث ل ث م ك ي د ل و ك ت ب ك ر م ل ط ع ت ل ا ح ي ف practical/regulatory, not school-curriculum history/geography
D27	Jordan primary GK: sheet metal screw definition — ه ت ي ا ه ن م بول سم لكش ل ا ي بول ول رام سم : وه جاصل ا نس ي غ ر ب vocational/practical knowledge